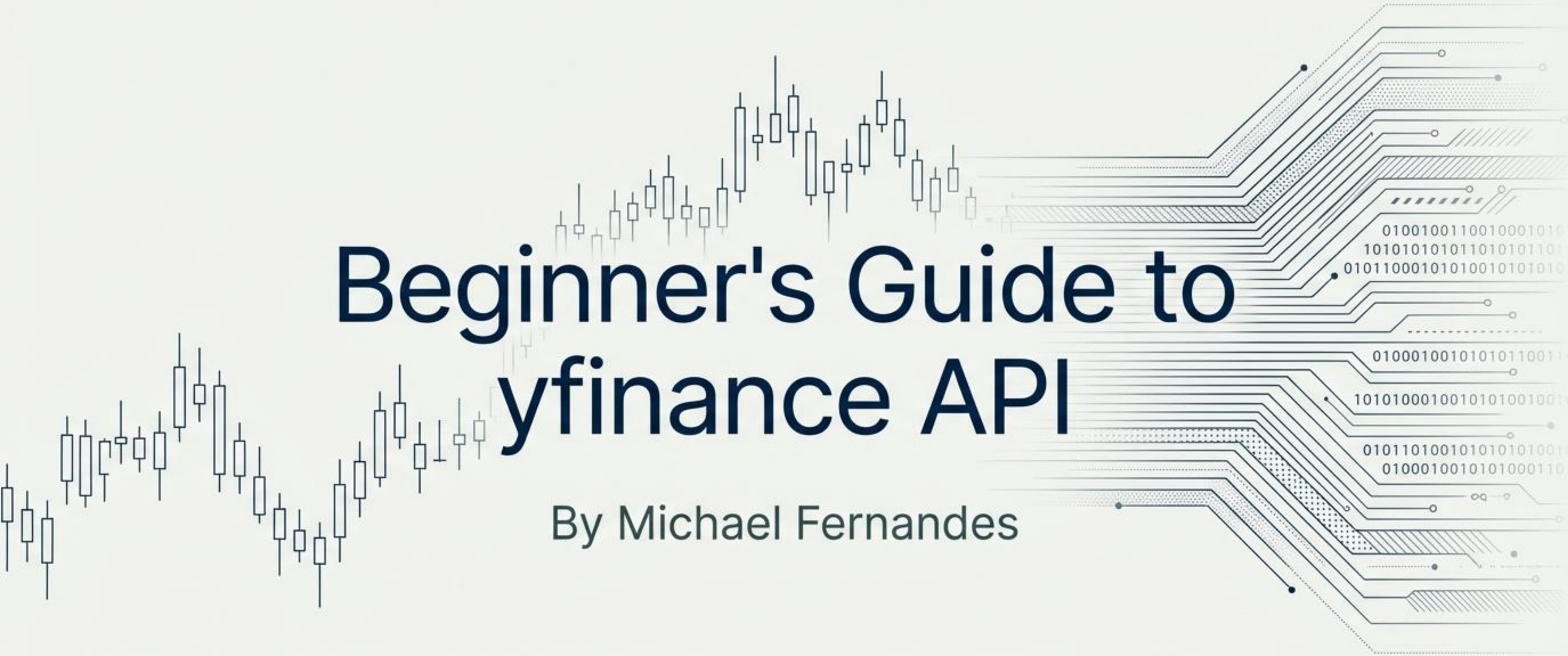


Beginner's Guide to yfinance API

By Michael Fernandes

Market Data with Python: The `yfinance` Cheat Sheet

Core Capabilities

Simple, Key-Free Setup

```
>_ pip install yfinance
```

Just `pip install yfinance` to start; no API key is needed.

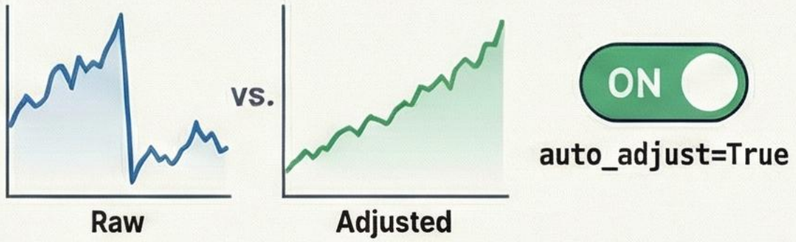
Fetch Historical Price Data

Use `ticker.history()` to get OHLCV data for stocks, crypto, and indices.

Access Company Fundamentals & More

Best Practices & Limitations

Always Use Adjusted Prices for Analysis



Set `auto_adjust=True` to account for dividends and splits in backtesting.

Be Aware of Common Pitfalls

Watch for missing data, rate limits, and survivorship bias.

A Research Sandbox, Not a Trading Engine

Ideal for **backtesting** and **prototyping**

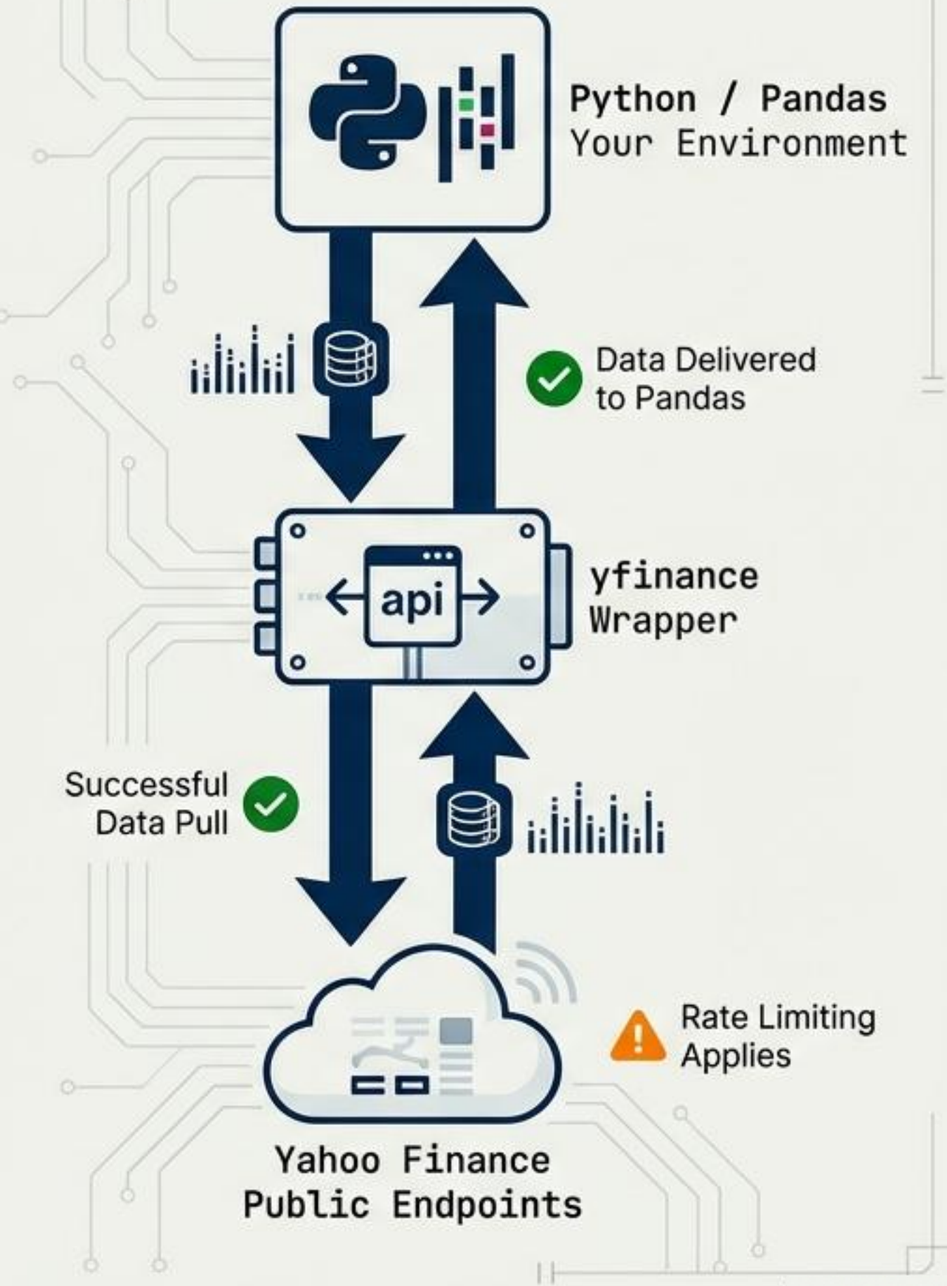
use **Bloomberg/Refinitiv** for **production systems.**

The Gateway to Market Data

yfinance is the most popular free Python API for pulling market data from public Yahoo Finance endpoints. It acts as a bridge for quants, students, and researchers to perform prototyping and exploratory analysis without expensive subscriptions.

The Research Sandbox

Ideal for strategy research, factor modeling, and academic projects. Not built for high-frequency trading or production execution.



The Foundation: Installation & Setup

```
$ pip install yfinance
```

script.py

```
1 import yfinance as yf
2
```

No API key required.
Immediate access.

The Core Object: The Ticker

Everything in the library revolves around the Ticker object. Instantiating this object connects you to the asset's data streams.

```
script.py
1 ticker = yf.Ticker('AAPL')
2
```

Universal Syntax

Stocks:
`AAPL`
`RELIANCE.NS`

Indices:
`^NSEI`
`^GSPC`

Crypto:
`BTC-USD`

Forex:
`USDINR=X`

Fetching History

The most common use case is retrieving Historical OHLCV prices.

```
# Basic History
df = ticker.history(period='1y')
```

Pandas DataFrame Output

Date	Open	High	Low	Close	Volume
2023-10-01	171.22	173.00	170.00	172.50	51234567
2023-10-02	172.50	174.10	171.80	173.75	49876543
2023-10-03	173.75	175.20	172.90	174.90	50112233
2023-10-04	174.90	176.00	173.50	175.80	48555666
...	...	...	...	...	...

Controlling Time: Intervals & Ranges

script.py

```
df = ticker.history(
    start='2022-01-01',
    end='2024-01-01',
    interval='1d'
)
```

Supported Intervals

Interval	Use Case
1m, 5m	Intraday (Limit: ~60 days history)
1h	Swing trading
1d	Standard strategies
1wk	Long-term analysis
1mo	Macro / Asset allocation

Decoding the Output: OHLCV

df.columns returns the standard financial data points.

Crucial for Quants: Adjusted Prices

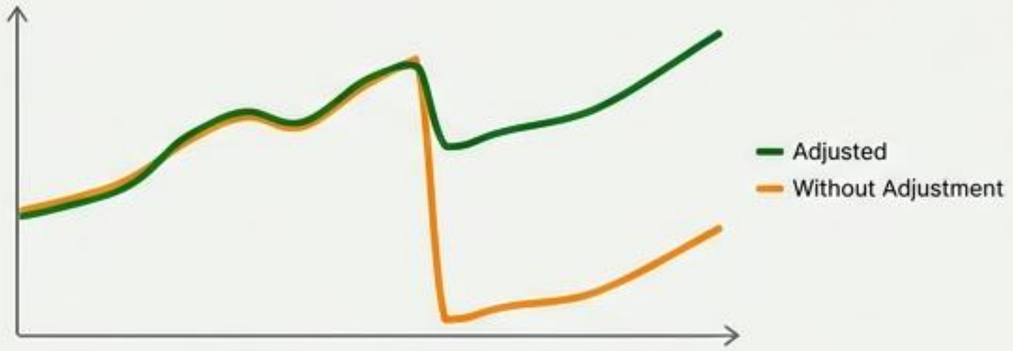
Warning: Unadjusted data invalidates backtesting.

script.py

```
df = yf.download('AAPL', auto_adjust=True)
```

This parameter automatically adjusts historical data for:

- Dividends
- Stock Splits



Scaling Up: Bulk Downloads

Efficiently retrieve data for portfolios or indices by passing a list of tickers.

script.py

```
tickers = ['AAPL', 'MSFT', 'GOOGL']
data = yf.download(tickers, start='2023-01-01')
```

```
data['Close']['AAPL']
```

data['Close']

AAPL	MSFT	GOOGL
175.50	93.00	99.02
175.50	93.50	99.07
175.50	98.00	90.80
AAPL	MSFT	GOOGL
175.50	99.30	99.73
175.50	95.80	99.03
175.50	98.50	90.20
175.50	99.30	99.50
175.50	95.80	90.80

Beyond Price: Company Fundamentals

script.py

```
# Annual Data
ticker.income_stmt
ticker.balance_sheet
ticker.cashflow

# Quarterly Data
ticker.quarterly_income_stmt
```

BALANCE SHEET	
Assets	
Total Current Assets:	$135,400 M
Cash & Equivalents:	$28,300 M
Total Non-Current Assets:	$245,100 M
Total Assets:	$380,500 M
Liabilities & Equity	
Total Current Liabilities:	$120,200 M
Total Non-Current Liabilities:	$150,500 M
Total Liabilities:	$270,700 M
Total Stockholders' Equity:	$109,800 M
Total Liabilities & Equity:	**$380,500 M**

Valuation & Metadata

The 'info' dictionary provides a snapshot of current valuation metrics and sector details.

```
info = ticker.info
```

marketCap

trailingPE

beta

sector

industry

dividendYield

Pro Tip: 'info' calls can be slow. Always cache these results locally.

Corporate Actions: Dividends & Splits

Track cash payouts and structural changes. Essential for Total Return calculations.

```
ticker.dividends
ticker.splits
```

Without accounting for these actions, price charts show false losses.

The Hidden Gem: Options Data

yfinance retrieves full option chains, including Volatility and Open Interest.

```
# Get expirations
ticker.options

# Load chain
opt = ticker.option_chain('2024-12-20')
calls = opt.calls
puts = opt.puts
```

Contract	Strike	Bid	Ask	Vol	Open Int	Implied Vol
XYZ241220C00150000	150.00	12.50	13.10	450	1200	**34.5%**
XYZ241220C00155000	155.00	9.20	9.80	300	850	**35.2%**
XYZ241220C00160000	160.00	6.50	7.00	220	600	**36.1%**
XYZ241220P00150000	150.00	5.30	5.80	180	450	**34.5%**
XYZ241220P00155000	155.00	7.90	8.40	210	520	**35.2%**
XYZ241220P00160000	160.00	11.10	11.70	260	680	**36.1%**

Asset Universality

Use the correct suffix/symbol format for global asset classes.

Indices

NIFTY 50 (^NSEI), S&P 500 (^GSPC)

Commodities

Gold (GC=F)

Crypto

Bitcoin (BTC-USD)

Forex

USD/INR (USDINR=X)

```
yf.download('BTC-USD', period='5y')
```

Technical Indicators (The DIY Approach)

yfinance provides raw data.
Use ‘pandas’ or ‘ta’ libraries to calculate indicators.


```
import ta

# RSI Calculation
df['rsi'] = ta.momentum.RSIIndicator(df['Close']).rsi()

# Simple Moving Average
df['sma_50'] = df['Close'].rolling(50).mean()
```

Critical Pitfalls (Safety Inspection)

Holidays
Data may be missing/null on market holidays.

Survivorship Bias - Delisted companies are often unavailable.

Rate Limits - Aggressive loops can result in IP blocks.

Intraday Instability - Intraday history is volatile and limited.

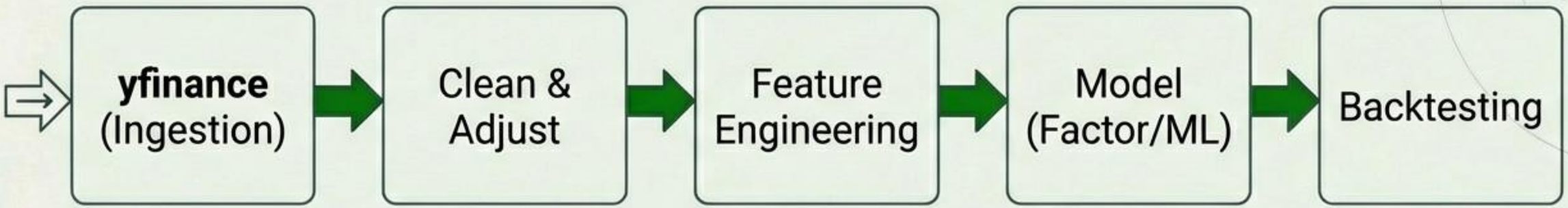
yfinance vs. The Giants

Feature	yfinance	Bloomberg/Refinitiv
Cost	Free	$$$$
Real-time	Delayed	Yes
Fundamentals	Inconsistent	Institutional Grade
Use Case	Research Sandbox	Production & Trading

Best Practices for Stability

- **Cache Locally** - Save to CSV/SQL to avoid re-downloading.
- **Validate** - Cross-check data points against secondary sources.
- **Research Only** - Do not hook directly into automated execution systems.

The Quant Workflow



yfinance is the starting point for modern Factor Investing and ML pipelines.